Video Processing.

Canny.

In an image, an edge is a place where the pixel intensity changes sharply.

Canny is an edge detection algorithm used to identify strong intensity gradients in an image, which correspond to object boundaries.

* Edge detections find locations where pixel values change rapidly, mathematically, it looks for high gradients.

Gradient = rate of intensity change.

Ex: If intensity changes slowly

$10 \rightarrow 12 \rightarrow 14 \rightarrow 15$

This is not an edge

If intensity jumps

$10 \rightarrow 10 \rightarrow 200$

It is an edge.

Simple visualization

→ original grayscale row

10 10 10 200 200

→ change b/w pixels

0 0 190 0

↑

Thus edge detected

Canny works in 5 main steps:

(i) Noise reduction
(ii) Gradient calculation
(iii) Non maximum suppression
(iv) Double threshold
(v) Edge tracking by hysteresis.

Consider i/p image (frame from a video)

$I(x, y)$

horizontal pixel location

vertical pixel location

$I(x, y)$ - intensity value.

whatever is the i/p image first converted to greyscale (0-255)

assume, $I = \begin{bmatrix} 10 & 10 & 10 & 200 & 200 \\ 10 & 10 & 10 & 200 & 200 \\ 10 & 10 & 10 & 200 & 200 \\ 10 & 10 & 10 & 200 & 200 \\ 10 & 10 & 10 & 200 & 200 \end{bmatrix}$

Step 1: Noise reduction

* Small noise can create fake edges. so we smooth the image using Gaussian filter.

i,e Original image → blurred image.

Mathematically, its convolution with a Gausian kernel.

i,e Gausian fn, $G(x,y) = \frac{1}{2\pi\sigma^2} e^{\frac{-x^2+y^2}{2\sigma^2}}$

Blurred image becomes = $I_s(x,y) = I(x,y) * G(x,y)$

convolution

Example kernel (3x3) at $\sigma = 1$

$$G = \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

for the center pixel we take this 3x3 region $\begin{bmatrix} 10 & 10 & 200 \\ 10 & 10 & 200 \\ 10 & 10 & 200 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} 10 & 10 & 200 \\ 10 & 10 & 200 \\ 10 & 10 & 200 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} / 16$$

$$= ((10\times1) + (10\times2) + (200\times1) + (10\times2) + (10\times4) + (200\times2) + (10\times1) + (2\times10) + (1\times200)) / 16$$

$$= 920/16$$

$$\approx 57$$

Similary for $\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$ after smoothning we get 10

for $\begin{bmatrix} 10 & 200 & 200 \\ 10 & 200 & 200 \\ 10 & 200 & 200 \end{bmatrix}$ after smoothning we get 152

So, o/p of this stage = $\underbrace{\begin{bmatrix} 10 & 57 & 152 \\ 10 & 57 & 152 \\ 10 & 57 & 152 \end{bmatrix}}_{\text{jump becomes smooth.}}$

<u>Step 2:</u> Gradient Calculation.

Here we calc how fast intensity changes.

we use sobel operator.

S_x (horizontal gradient) = $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

change left to right

$I_s \cdot S_x \Rightarrow \begin{bmatrix} 10 & 57 & 152 \\ 10 & 57 & 152 \\ 10 & 57 & 152 \end{bmatrix} \cdot \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$\underline{\underline{G_x = 568}}$

change top to bottom

S_y (Vertical intensity changes) $S_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

$I_s \cdot S_y = G_y$

$\Rightarrow \underline{\underline{G_y = 0}}$

This tells us $\Rightarrow$ strong horizontal gradient

$\therefore$ edge is vertical

Edge magnitude

$$G = \sqrt{G_x^2 + G_y^2}$$

$$G = 568$$

This G_x & G_y value is computed for each pixel in the image.

What we calc is only for the center pixel

edge direction

$$\theta = \tan^{-1}(G_y / G_x)$$

$$\theta = 0^\circ$$

i,e edge orientation is vertical.

Step 3: Non maximum suppression

Goal - making edges thinner.

* look at direction of gradient
* compare pixel with 2 neighbors in that direction
* keep pixel only if largest.

Let us assume we get

largest

$$G = \begin{bmatrix} 50 & 120 & 300 \\ 40 & 568 & 250 \\ 30 & 110 & 200 \end{bmatrix}$$

$$\text{Suppression results} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 568 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 4: Double Threshold

high threshold = 150

low threshold = 50

now we classify every pixel

Rules :

Value	Classification
> high	strong edge
low < value < high	weak edge
< low	not edge (0)

568 > 150 → strong edge rep as 255

<u>Step 5</u> : Edge Tracking by Hysteresis.

decide whether weak edges should be kept or removed.

Strong edges → always keep

weak edges → keep only if connected to strong ed

otherwise remove them.